EXISTENCE, UNIQUENESS AND UNIVERSAL PROPERTY OF THE DIVIDED POWER POLYNOMIAL EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. The universal-property check ends '(details omitted)': the omitted content is the mechanical multiplicativity verification on the divided power monomial basis; the construction and the inverse are displayed. The one-variable reduction is justified in-span (adjoin variables one by one).

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 1013 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Gluing lemma. lemma-two-ideals (same chapter, named). Builds the divided power structure on J from structures on IA<x> and A<x>_+ after checking the intersection condition.
- Extension along flat maps. lemma-gamma-extends (same chapter, named). Extends gamma to IA<x> since A -> A<x> is flat.
- Torsionfree criterion. lemma-silly part (4) (same chapter, named). Gives the divided power structure on A<x>_+ when A is torsionfree.
- Descent criterion. lemma-kernel criterion (2)(c) (same chapter, named). Handles the general (non-torsionfree) case via the R/a presentation of A.

Source reference. Original lemma 07H1: sources/stacks-dpa/pinned-original-source.tex, source lines 720–720. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07H2: sources/stacks-dpa/pinned-original-source.tex, source lines 825–825. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07GM: sources/stacks-dpa/pinned-original-source.tex, source lines 276–276. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07GQ: sources/stacks-dpa/pinned-original-source.tex, source lines 450–450. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let (A, I, γ) be a divided power ring. There exists a unique divided power structure δ on*

$$J = IA\langle x_1, \ldots, x_t\rangle + A\langle x_1, \ldots, x_t\rangle_+$$

such that

(1) *$\delta_n(x_i) = x_i^{[n]}$, and*
(2) *$(A, I, \gamma) \to (A\langle x_1, \ldots, x_t\rangle, J, \delta)$ is a homomorphism of divided power rings.*

Moreover, $(A\langle x_1, \ldots, x_t\rangle, J, \delta)$ has the following universal property: A homomorphism of divided power rings $\varphi : (A\langle x_1, \ldots, x_t\rangle, J, \delta) \to (C, K, \epsilon)$ is the same thing as a homomorphism of divided power rings $A \to C$ and elements $k_1, \ldots, k_t \in K$.

Proof. We will prove the lemma in case of a divided power polynomial algebra in one variable. The result for the general case can be argued in exactly the same way, or by noting that $A\langle x_1, \ldots, x_t\rangle$ is isomorphic to the ring obtained by adjoining the divided power variables $x_1, \ldots, x_t$ one by one.

Let $A\langle x\rangle_+$ be the ideal generated by $x, x^{[2]}, x^{[3]}, \ldots$. Note that $J = IA\langle x\rangle + A\langle x\rangle_+$ and that

$$IA\langle x\rangle \cap A\langle x\rangle_+ = IA\langle x\rangle \cdot A\langle x\rangle_+$$

Hence by Lemma 07GQ it suffices to show that there exist divided power structures on the ideals $IA\langle x\rangle$ and $A\langle x\rangle_+$. The existence of the first follows from Lemma 07H1 as $A \to A\langle x\rangle$ is flat. For the second, note that if A is torsion free, then we can apply Lemma 07GM (4) to see that δ exists. Namely, choosing as generators the elements $x^{[m]}$ we see that $(x^{[m]})^n = \frac{(nm)!}{(m!)^n} x^{[nm]}$ and $n!$ divides the integer $\frac{(nm)!}{(m!)^n}$. In general write $A = R/\mathfrak{a}$ for some torsion free ring R (e.g., a polynomial ring over $\mathbf{Z}$). The kernel of $R\langle x\rangle \to A\langle x\rangle$ is $\bigoplus \mathfrak{a} x^{[m]}$. Applying criterion (2)(c) of Lemma 07H2 we see that the divided power structure on $R\langle x\rangle_+$ extends to $A\langle x\rangle$ as desired.

Proof of the universal property. Given a homomorphism $\varphi : A \to C$ of divided power rings and $k_1, \ldots, k_t \in K$ we consider

$$A\langle x_1, \ldots, x_t\rangle \to C, \quad x_1^{[n_1]} \ldots x_t^{[n_t]} \longmapsto \epsilon_{n_1}(k_1) \ldots \epsilon_{n_t}(k_t)$$

using φ on coefficients. The only thing to check is that this is an A-algebra homomorphism (details omitted). The inverse construction is clear. □